LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	31
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Protection and location of flight data recorder
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Protection and location of flight data recorder.
5. Specific Objectives	15.1.11.4 Trainee will be able to explain Protection and location of FDR.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of FDR installation diagrams in IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the Flight Data Recorder?
(b) Where is the FDR typically located on a commercial aircraft?
(c) Why is the FDR often painted orange?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• FDR physical protection mechanisms • Crash survivability requirements • Environmental sealing standards • Impact resistance specifications	L & Ds	PPT, IB	15 min
(b)	• Standard FDR installation locations • Tail cone mounting procedures • Fire protection systems • Access panel configurations	L & Ds	PPT, PDF	20 min
(c)	• Regulatory compliance for FDR • Maintenance access protocols • Data integrity safeguards • Post-accident recovery procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the FDR protects data during a crash event.
(b) Describe the typical location of the FDR relative to the aircraft center of gravity.
(c) What measures ensure the FDR remains operational after fire exposure?

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of FDR protection and location will assist in future troubleshooting of aircraft black box systems during LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Direct reading magnetic compasses- Magnetic properties, hard iron and soft iron, terrestrial magnetism, magnetic variation, magnetic dip.	Refer to the master syllabus reference list	LP 32

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min